# Trabalho Calculo -

① a) $(\ln x + 5^x)^9$

$$9(\ln x + 5^x)^8 \cdot \left(\frac{1}{x} + 5^x \ln 5\right)$$

b) $\operatorname{sen}(x^3 + \ln x)$

$$\cos(x^3 + \ln x) \cdot \left(3x^2 + \frac{1}{x}\right)$$

c) $2^{\cos x}$

$$2^{\cos x} \ln 2 \cdot (-\operatorname{sen})$$

d) $\sqrt{3x^2+4}$

$$(3x^2+4)^{1/2}$$

$$\frac{1}{2}(3x^2+4)^{-1/2} \cdot (3.2x + 0)$$

$$\frac{1}{2}(3x^2+4)^{-1/2} \cdot (6x)$$

② a) $2x^3 + 4y^2 = 12xy - 20$ $\boxed{P = (2,3)}$

$$2.3x^2 + 4.2yy' = 12y + 12xy' - 0$$

$$6x^2 + 8yy' = 12y + 12xy'$$

$$8yy' - 12xy' = 12y - 6x^2$$

$$y'(8y - 12x) = 12y - 6x^2$$

$$y' = \frac{12y - 6x^2}{8y - 12x}$$

$$12.(3) - 6.(2)^2$$

$$36 - 24 = 12$$

$$8.(3) - 12.(2)$$

$$24 - 24 = 0$$

$$y' = \frac{12}{0} = \infty$$

1. b) $7y + y^4 - x^3 = -23$ $\boxed{P = (5,3)}$

$$7y' + 4y^3 y' - 3x^2 = 0$$

$$y'(7 + 4y^3) = 3x^2$$

$$y' = \frac{3x^2}{7 + 4y^3}$$

$$3.(5)^2 = 75$$

$7 + 4 \cdot (3)^3$

$7 + 108 = 115$

$y' = \frac{75}{115} = \frac{15}{23}$

$y - 3 = \frac{15}{23}(x - 5)$

$y = \frac{15}{23}x - \frac{75}{23} + 3$

$y = \frac{15}{23}x - \frac{6}{23}$

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3) $A = (500, 50t)$ $\quad 22 - 12 = 10h$

$B = (0, -70t)$

$d(t) = \sqrt{(500-0)^2 + (50t - (-70t))^2} = \sqrt{(500)^2 + (120t)^2}$

$d(t) = \sqrt{250000 + 14400t^2}$

$\frac{dd}{dt} = \frac{1}{2}(250000 + 14400t^2)^{-1/2} \cdot 28800t$

$$\frac{dd}{dt} = \frac{14400\,t}{\sqrt{250000+14400t^2}}$$

$$14400 \cdot (10) = 144000$$

$$\sqrt{250000+14400\cdot(10)^2}$$

$$\sqrt{250000+1440000}$$

$$\sqrt{1690000} \simeq 1300$$

$$\frac{144000}{1300} \simeq 111\ km/h$$

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④ a) $\lim\limits_{x \to 1} \dfrac{4x^3-5x+3}{x \ln x}$

$$\begin{cases} 4x^3-5x+3=2 \\ x\ln x = 0 \end{cases}$$

$$\frac{2}{0}$$

Se $x>1$

$$\begin{cases} 4x^3-5x+3>0 \\ x\ln x > 0 \end{cases} = +\infty$$

Se $x<1$

$$\begin{cases} 4x^3-5x+3>0 \\ x\ln x < 0 \end{cases} = -\infty$$

b) $\lim_{x \to -\infty} \frac{9x-7}{3x+e^x}$

$$\frac{9x-7}{3x+0} = \frac{9x-7}{3x} = 3 - \frac{1}{3x}$$

$$3 - 0 = 3$$

c) $\lim_{x \to 0^+} x^4 \ln x$

$$\frac{\ln x}{x^{-4}} = \frac{-\infty}{+\infty} \qquad \frac{\frac{1}{x}}{-4x^{-5}} = \frac{-x^4}{4}$$

$$\lim_{x \to 0^+} \frac{-x^4}{4} = 0$$

d) $\lim_{x \to 0^+} (1+7\,\text{sen}\,x)^{1/5x}$

$$\frac{\ln(1+7\,\text{sen}\,x)}{5x} = \frac{0}{0} \qquad \frac{\frac{7\cos x}{1+7\,\text{sen}\,x}}{5} = \frac{7\cos x}{5(1+7\,\text{sen}\,x)}$$

$$\lim_{x \to 0^+} \frac{7\cos x}{5(1+7\,\text{sen}\,x)} = \frac{7}{5}$$

⑤ $x^4 - 8x^3 + 18x^2 + 9$

$f'(x) = 4x^3 - 8.3x^2 + 18.2x + 0$

$= 4x^3 - 24x^2 + 36x$

$= 4x(x^2 - 6x + 9)$

$= 4x(x-3)^2$

$4x = 0$ $\qquad$ $(x-3)^2 = 0$

$x = 0$ $\qquad$ $x = 3$

| | | | |
|---|---|---|---|
| $4x$ | $-$ | $+$ | $+$ |
| $(x-3)^2$ | $+$ | $+$ | $+$ |
| $f(x)$ | $-$ | $+$ | $+$ |
| | | $0$ | $3$ |

$f''(x) = 4.3x^2 - 24.2x + 36$

$= 12x^2 - 48x + 36$

$= 12(x^2 - 4x + 3)$

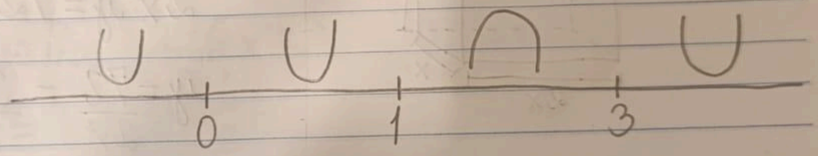
$= 12(x-1)(x-3)$

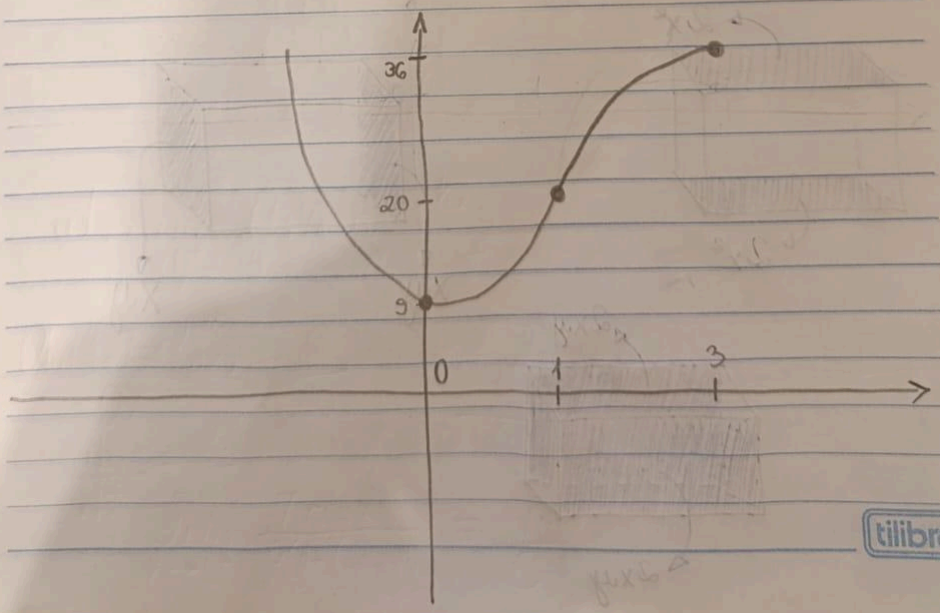
$(x-1)=0$ $\qquad$ $(x-3)=0$

$x=1$ $\qquad$ $x=3$

| | | | |
|---|---|---|---|
| $(x-1)$ | $-$ | $+$ | $+$ |
| $(x-3)$ | $-$ | $-$ | $+$ |
| $f(x)$ | $+$ | $-$ | $+$ |

1 $\qquad$ 3



Máximos locais → Não

Máximos globais → Não

Mínimos locais → Sim (0,9)

Mínimos globais → Sim (0,9)

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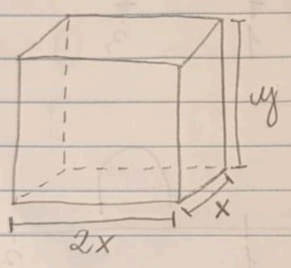
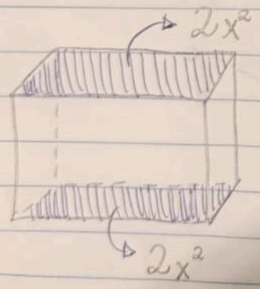
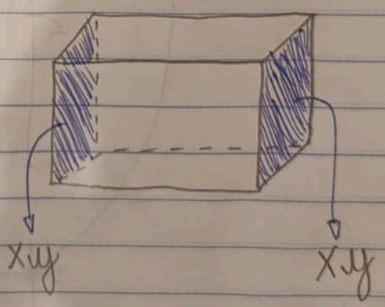


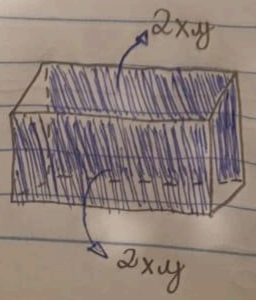

$$2x \cdot x \cdot y = 72$$

$$2x^2 y = 72$$

$$y = \frac{72}{2x^2} = \frac{36}{x^2}$$



$$Q = 4x^2 + 4xy + 2xy$$

$$= 4x^2 + 6xy$$

$$4x^2 + 6x \cdot \frac{36}{x^2}$$

$$Q(x) = 4x^2 + \frac{216}{x} = 4x^2 + 216x^{-1}$$

$$Q'(x) = 8x + 216 \cdot -1x^{-2}$$

$$= 8x - 216x^{-2}$$

$$= 8x - \frac{216}{x^2}$$

$$8x - \frac{216}{x^2} = 0$$

$$8x = \frac{216}{x^2}$$

$$x = 3$$

$$8x^3 = 216$$

$$x^3 = \frac{216}{8} = 27$$

$$2x = 2.(3) = 6\ m$$

$$x = 3\ m$$

$$y = \frac{36}{(3)^2} = \frac{36}{9} = 4\ m$$